

ActInf Livestream #004.2: "Cultural Affordances: Scaffolding Local Worlds"

Livestream | 1:36:43

all right i think we are live
hello everyone and welcome to team com
podcast 4.2
this is an exciting day because we are
testing out new technology
and so thanks to all of our participants
we hope this works for all of you
listening and in attendance
let's go to the slides welcome to
teamcomp teamcom is an experiment in
online team communication
and learning related to active inference
you can find us on twitter at our
inferenceactivehandle
you can email us at activeinference
gmail.com
find us at our public key based team or
watch us on youtube
this is a recorded and archived live
stream
so please provide us with feedback so we
can improve on our work
all backgrounds and perspectives are
welcome here we welcome
all questions that can help people
understand active inference
and as far as basic video etiquette
please mute if it's noisy in your
background
feel free to use the raise hand feature
in jitsi so that we can
make sure to call on you we are here in
active inference stream 4.2 following up
on 4.1
in 4.1 we discussed cultural affordances

scaffolding local worlds through shared
intentionality and regimes of attention
a 2016 paper by ramstead at all
today we're going to do a check-in and
warm-up
and last time we had a discussion of the
daniel
we can see slice by now
yes okay good point i can see the slides
on the live stream
but uh they're not being shared in the
jitsi
so what i'm going to do and we're um
doing it live i'm going to post the
live stream link and if you'd like you
could have the live stream
muted in the background and just see
what the slides are because it will just
be rapidly
um following so i'll continue on this
first part and
the link will be dream
put into the jitsi chat if you could
just give me
one second how about something on active
inference and multitasking
that's what i want to know about
last time we had an awesome discussion
of the paper
we went through the abstract the claims
and the aims of the work
and we also went through all the figures
in pretty nice detail
this time we are going to have further
discussion which has been separated into
a few different topics that
none of the conversions have been
prepared on beforehand so it will be

fun and improvisational so let's start
with the check-in
and warm-up people can introduce
themselves and their location if they'd
like
the first question is what is a local
world that you live in
any thoughts on local worlds
it's a good question um
i mean i i you know i guess this says
something about the way that
we live i mean you know i i live in
montreal
and my local world uh i live around the
mcgill
area so it's very studenty um
i guess i i live in a world where
people share progressive values and i
also get the feeling that there's
something like
a rift you know where where
several of us are sharing the same
geographical location um but don't
necessarily live in the same world
uh per se in the same in the sense of
the things don't afford the same kinds
of things
that they do uh it you know for
for other people living in another as it
were world
um yeah yep the first thing i thought of
was that local
can be geospatial or it can be
informational
one informational link away is just one
video chat away one phone call away
and what we're experiencing in a big way
is that tension between our

informational local worlds and our
spatial ones and
that can lead to um discordance
any other thoughts on this part
cool maybe uh
in this term of local maybe it's
based on communications which happens
most often with you
if even in a global environment but
if you communicate each day with
somebody
your kind of local world
cool second question what gives meaning
to our local worlds so this can be maybe
a thought from the paper
what did they propose gives meaning to
local worlds or just
experientially what is it that gives
meaning to your local world
yeah stephen well i think there's the
geographical
like physical embeddedness of that world
so how you actually when i think of
world i kind of think
very much of it in terms of the physical
space
being part of the of that whole piece
um but then like you say how those
things show up
then starts to become maybe more
abstract or more
inferred through other people
yep and from a process perspective as
we'll
get to later it's the feedback loops
that happen within ourself and amongst
various people that provide
semiotic meaning sasha

yeah for some reason i just thought of
this phrase like theory of place
kind of going along with theory of mind
that in this kind of
uh local information environment but
maybe not
geospatially local that the people that
you're communicating
with you may or may not know where
they're
communicating from and that um
could be helpful in your communication
or not because um
simply because we're all here on this
live stream we don't really need to know
the exact location of each of the
participants
but if i did know it might add something
to our communication
or the narrative that we're building
to kind of know where we can all connect
from
and then to follow up with the thinking
through other minds
idea it's like thinking through other
spaces it's about putting yourself
in someone else's shoes understanding
where they're coming from
these are embodied and spatial metaphors
that we use
just like can you grasp it can you
sketch out the idea
these are tactile ways to talk about
abstract things like ideas
and so you're literally putting yourself
in someone else's shoes if you ask
what is the accessibility of this room
going to be for somebody who's coming

from a different place
and these are the important things to
consider
and we'll return to this i think when we
talk about the third space that exists
in social interactions
um in terms of the uh the paper cultural
affordances the
proposal there is something like well
um what what's at stake is something
like a dialectic
between affordances and expectations
right so
uh what what ends up shaping the
the worlds the that we inhabit right
which which we're conceptualizing in
terms of affordances
um really are the the shared
expectations that we have uh you know
with with other people in our community
and by shared expectations i mean
something very broad
uh to encompass uh the beliefs that we
have about
uh the shared world but also the norms
uh that are
kind of structuring uh our appraisal and
interaction with
the world as well as practices that
might be ongoing
and crucially the the shared physical
environment
that we have in common that in in the
case of humans
comes to encode information about the
way that we live
so you know think about uh the fact that
in most cities

streets meet at right angles uh or you
know the
use of circular uh traffic signalization
uh to kind of coordinate our patterns of
movement throughout a city
i think what's at stake is really the
this dialectic
yep the ant colonies architecture is in
feedback with their behavior
and they're exactly updated and
generated by the very
same process which is nice construction
or distributed cognition
different contexts are going to have
different terms uh final warm-up
question
is what are you trying to get a grip on
this is looking forward to talking about
grip later on
in this discussion but what's just
something that you're trying to get a
grip on i just thought it'd be a fun
random question
one thing we were talking about just
right before we started was trying to
get a grip
on some of these formalisms and
understanding which formalisms
in the free energy principle and in
active inference are
downstream of each other which ones are
necessary or sufficient for each other
and also what is the best way to on-ramp
and to learn about them
from various different perspectives
i was going to say exactly that i was
going to talk about the reading group
and the progress that we're making there

uh yeah what have you found interesting
or liked about that
well i mean i didn't expect this to
really turn into
the rich discussion and exchange
community that it's turned into uh
i mean uh yeah we're really coming to
grips i think as a group with
uh with the basically all the math and
physics
that goes into it like really uh
becoming familiar with all the moving
parts and all the ingredients it's been
a really
positive experience so that's what we're
i guess i'm also trying to learn
that guitar a bit more that's another
thing i'm coming to literally
grips with but those are the two main
things i'd say nice and stephen do you
raise your hand
yeah i like well like you were so
talking about this idea of putting
yourself or getting in someone else's
shoes
so i'm trying to like put myself
or trying to understand to put myself in
someone someone else's shoes in terms of
understanding what i'm talking about
this stuff and then
i think the whole idea of like am i
trying to understand
um in this dialectic with someone else
where they are
in relation to me or am i actually
getting into their shoes
in a place or in a field of practice
or in some sort of stance in relation to

um some concepts so i think there's
something kind of about
actually how we try and understand what
it means
in a meta level like you know because
and how we move between them because we
i tend to find that a lot of
um the stuff like anthropology and that
it tends to have
particular metaphors that we end up all
having to follow
from gold or whoever and that's
particularly true in the arts
it's all like it's all to lose and to
lose and
lose and then everything's those
metaphors
so like how how do you jump in and out
of those
different sort of shoes that people are
in
i think that's really interesting yep
being specific
with generalizations being tangible with
the abstraction
those are tensions that we work with and
also one thought on local worlds and
getting a grip
so once there's a grip on the object it
becomes part of your
extended cognition if you have the grip
on the hammer you can hit the nail but
if you don't have a grip on the hammer
it's not part of your
affordances to nail that in with a
hammer or if you're in someone else's
shoes
the local world kind of becomes the foot

shoe
container and the local world is what
gives meaning to that
coupling between the foot and the shoe
between the hand and the hammer
and so in in just a spatial way but
perhaps also in an abstract way
the local world is tying back to the
affordance idea and upon that control
um idea sasha
sasha muted oh uh yeah just to add a
little uh you know metal level
uh comment on that um the way that we
communicate um right it
has to capture the content of what we're
trying to communicate
and then also in a way that makes sense
to
uh the receiving party but it's just
funny especially
on this sort of podcast conversation
style that um
we're trying to communicate with each
other but also in a way that is going to
be
understandable to a future audience
and so i think there's a few different
levels of modulating
um our communication style and
um the kind of affordances that we have
perfect transition to our first bucket
of questions which is
feedback and affordances so on this one
this slide has a picture of a termite
nest
it also has a picture of some brain
computer
images and there's a quote from the

paper it reads

this multi-level embodiment of the

generative model

as we shall argue below extends to the

concrete material

human design milieus or designer

environments in which humans operate

some generative models in this wide

sense involve semantic content and

others do not

they involve something more minimal than

satisfaction conditions

i.e reliable covariation so

my question which is pretty broad was

what does active inference

tell us about niche construction in the

physical sense or in the cultural sense

there's a lot to say there and there

were many subsequent papers

um well i mean if i can just uh make

a few points uh

to get us started um so

axel constant and yellow brinaberg i

think are the two

first people to really start working on

the niche construction aspect

of active inference back in 2017 or so

and so all this work rests on the fact

that whenever we're talking about

active inference we're talking about a

story that's symmetric

in some deep sense so active inference

is a story about how

uh organisms over the course of uh

you know development and uh

and learning and through perception and

processes like evolution

become more like their ecological niche

which

technically means that their statistical

structure comes to recapitulate

uh in some sense the structure of the

environment

uh in which uh the you know the

organisms that we're interested in

exist um so for example uh

you know you can tell a lot about my

environment just by looking at my

physical body right

so the fact that i'm wearing a parka

tells you that summer's over in canada

for example so there are these kind of

predictive relations between the state

of my phenotype

and the expected state of the

environment to kind of come together

so the other side of active inference

that's often emphasizes the active bit

so you know rather than making my

beliefs more like the environment i can

make my environment more like my beliefs

or

conformed to my preferred data

distributions

what was overlooked before 2017 is that

from the point of view of the organism

and environment the story is symmetric

so when we're making the environment

more like our preferred data

distributions

as an active inference we're effectively

changing the structure of enviro of the

environment

to to make it more like our own

structure or to make it like

the the like the structure of the the

preferences that we're encoding
um and so uh i think a lot of
a lot of the more interesting work done
on culture
recently in the active inference
tradition draws
on uh this niche construction stuff i
mean you know at some point the idea is
you've only got so many layers of cortex
and cortex cortex is metabolically
expensive
so where are you gonna keep optimizing
you know after like the sixth layer
or whatever of like increasingly meta
increasingly centralized uh
representation well one way to go is
really just in the environment
so there's a there's a sense in which
really like coordinating
amongst amongst con specifics you know
like uh
or or with a team member or whatever is
sort of like
constructing a niche in another person
in a sense you're sort of like
you know uploading into another person
the sets of priors that will optimally
allow you to interface together uh which
relates to some of the top top effects
that we were discussing last week
so i mean ultimately the picture that i
kind of draw
for multi-scale active inference is that
you have two stacks
essentially you've got a kind of
vertical stack and a horizontal stack
by vertical stack i mean that all the
systems that we're considering are

nested systems of systems of systems

right so

uh you know our bodies are made of

organs that are made of cells and so on

um or you know uh the broader uh

socio-economic uh community in north

america is made up of like you know

more regional groups like the u.s canada

mexico

which in turn are made up of you know so

you got these kind of nested

uh structures going and

the yeah the active inference

story then can be used to

i think make sense of the ways in which

we structure the environment especially

humans this is

especially the case for humans that we

structure the environment in order to

foster

adaptive patterns of group behavior

any other thoughts on that and i can add

as well

well and raise your hand if so one part

that i thought was really interesting

maxwell was this notion of like a deep

symmetry

especially when the two interacting

components of the system have very

different affordances

so i was thinking about winter just like

you brought up

we don't recapitulate uh snow or

cold weathers however we do come to

embody

those regularities in the day and in the

year

with our first our thermoregulation as

an organism and then
with our behavior and with our extended
phenotype with our buildings and with
our air conditioners
and so we do in a very deep sense come
to embody those regularities
and then the air conditioners change the
climate in some
sense maybe a little maybe a lot maybe
it's in a good direction maybe it's in a
bad direction
but there is a consequence of this
coming to embody the regularities
and especially when the systems are very
different maybe when you're doing a
call and response it looks symmetrical
at the syntactic level and it has this
deep symmetry at the turn taking level
and then with other systems where the
where the turn taking looks very
different
it's still something symmetrical but
it's not quite
symmetrical at the first pass right well
so the the uh the idea was i was trying
to
uh elaborate a bit earlier with this
vertical and horizontal stack is
touches precisely on this is that it's
not just that
organisms are stacked in this kind of
vertical
ways that you also have this kind of
horizontal relationship between an
organism
or an agent and its niche that turn out
to also be stacked in this kind of way
so it's not just that my cells make up

organs that make up organisms
that make up my body for instance it's
that my body is a niche for or
for my organs and my organs are a niche
for my cells
and so when we're talking about niche
construction it's very much as sasha was
describing a few a few moments ago it's
that we're thinking
through the environment that is
populated by others we're thinking
through other worlds
uh by the way i noticed that my camera
is very ours is not so you can
reload one one thought i'll give there
on the
uh the primacy of the organism from a
philosophy
of biology or natural selection
perspective is
it was seen that the organism or more
specifically to some
the genes of the organism are the only
thing that make it through the weismann
barrier the only thing to make it
into the next generation is the
information that encodes the organism
in one sense and so that says okay well
the organism is the unit of analysis and
things lower our
subgroups yes the beta cells in the in
the pancreas have a niche that's
surrounded by other types of cells and
yes the oocyte is supported by other
kinds of cells but
still the organism is the unit of
selection and so that gives
um some people a reason to think that

that
is the default and that anything bigger
than an organism is collective behavior
anything smaller than the organism is
molecular physiology or some different
field
but some of the implications of taking
this multi-scale proposal
seriously are that we still get the
richness of collective behavior
of lateral interactions within a level
as well as the hierarchical
layering and that combination of
hierarchical layering
with lateral interactions um goes by
various names including a heteroarchitectural
system from the social insect literature
and
when we start thinking about things like
the whole genome like the genome
encoding a developmental trajectory
in relationship with an environment and
a microbiome
pretty interesting stuff and it leads to
question what the unit of selection is
is it the organism is it the kin group
is it the kin group and the extended
phenotype and
one of the ways that we at least delay
the question
while deepening our understanding is by
just observing
and by modeling in the most integrative
framework that we can imagine
and then many of these other things fall
out many of the relationships between
higher and lower levels fall out but
just because there's a relationship

between a higher and a lower level
spatially
or laterally doesn't mean that one level
or the other is the one where things are
actually happening it's actually
happening across all the levels
okay oh yeah i mean i think that there's
a there's a perspective that's being
opened up here that really allows us to
move beyond
you know these kind of simplistic and
empirically inadequate models like homo
economicus
like this idea that what human beings
are really doing at the end of the day
is optimizing their individual expected
utility
uh when they're performing tasks i mean
it
it turns out that humans act in a way
that's
deeply social we events a fairness
psychology
not nash equilibrium uh type so there's
an interesting work in comparative
ethology that puts
uh chimpanzees against humans in a
variety of
economic tasks like the ultimatum game
and yeah it turns out that humans just
don't operate
according to the principles of classical
economics uh
we don't zero in on nash equilibria it
turns out that classical economics is
actually pretty good at predicting the
behavior of chimps though
and so uh it's probably the case that

humans in particular are subject to
forms of social
and cultural selection like forms of
like
let's let's use the word you know it's
group selection right
and you know this might have been spooky
and hard to explain a while back but
with the nested markov blanket formalism
i think we're finally getting to a point
where we can talk
about group selection in a way that's
not just slippery
uh and in a way that doesn't reduce
everything being selected
to as you're saying dan uh to the genes
yeah
um two quick points on that first with
the homo economicus
it debases us twice it says your
behavior is fundamentally economic
one strike and two you're not even that
good at it you make errors or you have
these
cognitive biases that lead you to make
false positive investments or false
negative
instead we could just take a step back
and observe
how we see people interacting in a
phylogenetic context just like you were
describing
and maybe that is how we are which is
that we are
pro-social and it's not an error
correction on an error correction
on a debasement it's actually we can
step into the richness of us as

social creatures rather than thinking of
sociality as being recovered
after we've already you know paid off
everybody um and then also
the question about group and kin
selection and all these different
levels of selection debate first off in
the
last several years through the work of
akasha and others
it has been described that some of these
mathematical equations
are actually formally identical so
there's an isomorphism and it's like you
can
model it different ways but you're gonna
get to some of the same consequences
um and we can think about the
affordances of a single cellular
bacteria let's say now a single cellular
bacteria's niche
may also involve metabolites from con
specifics and social interactions like
quorum sensing
um that's an example where the
replicator is still the single bacteria
however there's collective behavior like
quorum sensing and then
the major transitions in evolutionary
individuality
so like the prokaryote plus another uh
something to make a eukaryote eukaryotes
coming together to make multicellular
organisms and
multicellular insects becoming the
eusocial organisms
it's not based upon you know formica
economicus it's not the ants making an

economic trade-off
it's that when their affordances change
in the context of a niche
you end up seeing an evolutionary
trajectory that
favors obligate cooperation at that
scale
and precisely precisely it sidesteps a
lot of these
how would i feel if i were an aunt who
didn't get to reproduce
type questions which are really uh
sidetracks
relative to what is happening yeah and
you know i think uh
it's doing a lot of damage if we're
being real about it i mean you know the
these these economic models are not just
flawed
theoretically and empirically inadequate
they're also
driving investment um and you know the
the current state of the economy i think
which is you know despite what you might
hear from propaganda like not not so
great right now i mean
it's uh it it i think in part it has
to do with the
flawed way in which we're understanding
you know
human economic behavior what we really
need to be doing is fostering a move
towards like you know
value or affordance-based economies yeah
or instead of how can we give everybody
enough
u.s coin so that they can buy food
another way to phrase that is

how could we reduce people's uncertainty
about their next meal
and maybe sometimes it looks this way
sometimes it looks that way
but if we were focused on reducing
physiological surprise
rather than as a proxy trying to
increase some value on a spreadsheet
thinking that people would then
um just acquire the foods it's an
interesting
idea if we pivoted from these
reward-based framings of political or
economic issues to
uncertainty based issues because it's
often what you hear people
needing help with is not knowing where
the next place they're going to sleep or
get their meal from
not i don't have enough to get a meal i
don't know where the meal is coming from
so it's actually interestingly an
uncertainty question even in that case
oh yeah absolutely here's our next slide
another quote from the paper is we
hypothesize
that the central mechanisms whereby
cultural affordances are acquired
especially conventional context
involving affordances
consists in looping our feedback
relations between shared intentionality
and shared attention in this light one
can view social norms and conventions as
devices to reduce mutual uncertainty
that is constantly with a free energy
framework as entropy minimizing devices
one must know what is in the minds of

others such as

what one would see and how one would
interpret another's action generally and
in context

in order to make a successful inference

both explicit

content involving or implicit

correlational inferences about other

agents in each simulation situation

sorry so the question here was

how do narratives reduce uncertainty how

does conversation

that's a great question um i mean you

know

you all did uh our paper on narratives a

few weeks ago so i'm sure you have some

elements so to answer that already but i

mean that

the fundamental idea i think uh is the

and also i don't want to take up too

much room if other people have

more interesting things to say please uh

by all means

you know jump in but i think you know

narratives do this in

two main ways uh following the two main

senses of narrative that we've outlined

right so

by a narrative you might understand uh

some kind of internal cognitive shama

right so like when you tell tell the

story of your life or whatever

and like from that point of view what

narratives allow is

for a kind of a novel

kind of useful uh re-representation of

lived events that you've already

undergone so it's a way of kind of

introducing new structure and meaning
into the flow of events that that have
happened to you
but by narrative you might also mean a
shared
story so literally a cultural artifact
that is produced through a storytelling
practice
and uh you know from from a niche
construction point of view
these are shared elements of a common
environment
that we interact with and effectively uh
internalize the structure of such that
we're we're then able to uh you know
redeploy the ideas
concepts values and so on that are
present
in the structure of that narrative to
structure our social life
if i had in a nutshell i think those are
the two main functions
yep um and please anyone on team com
raise your hand if you want to bring in
the paper uh the idea of that
narrative as almost being embodied in
our evolutionary priors reminds me of
the hero's journey
and the the narrative structure of the
hero's journey it's not about
the hero tying and untying their shoes
it's about a story arc which goes from a
to z and that does many things it
compresses the past
into a distilled representation so that
you don't need to remember
every single detail of the past but
rather you can

remember the action-relevant sections of
the past

like the bird told me something relevant
you don't need to remember how many
feathers were on the bird

it's enough to know that in a certain
space and time you had an interaction
where the bird told you go down this
path not that one and then the symbolism
as well and so words are symbols these
are things we'll return to as well

and so the narratives are like paths of
symbols whether strings of words
um or evocative uh patterns

yeah stephen

yeah and this also probably changes a
bit in terms of how indigenous cultures
work with story and legend
and animalism and
unknowns in the environment which they
were tuning to

and the way that we've kind of codified
our narratives these kind of grand
narratives which

rather than it being about the land
which indigenous culture tends to be
about the land

and timeless qualities that are
enfolding

you get this kind of you know you
mentioned about the past as an
entity that's in a timeline

where we have a sense of self in the
timeline which is actually
kind of a more modern way of thinking
about things so

in some ways we've actually designed an
understanding of how we fit into the

world
because our niche has changed in the way
that
you know you were talking about because
we've constructed
our own way of understanding what
our sort of human world is and attune to
that
now we're kind of we're even redefining
the way
we see our history and the way we see
narratives because narratives are these
broader temporal depths which can
stretch
and then stories have become kind of
commodified whereas in indigenous
culture
and story and land and us
are much more kind of in the present
in this kind of unfolded kind of cycle
that
nature provides rather than saying what
our calendars
or our watches provide yeah two
thoughts on that one is the distinction
between kronos and kairos
so the chronometer is the clock that's
tick tick tick tick tick
that's the regularities and the decimal
system and kairos means timeliness
and so it relates to dinner time versus
6 p.m those are very different things
one is an exact time independent of an
embodied experience
and the other one is actually the
experience independent of the clock time
so that's important and then also just
to highlight um the pragmatism

of taking an action-oriented approach or
the active inference approach to
thinking about our modern life
as team com we often find it helpful
when beginning and closing a meeting
like bookends
to almost have narrative review just
restating syntactically here's who's
here
here's what we intend to get out of this
interaction or
here are the goals of the meeting in
previous
frameworks like getting things done is
technically an action-oriented framework
but not in the inactive embodied sense
it's sort of in a productivity
homo economicus framework and so you get
convergent evolution
of sometimes the recommendations like
make a to-do list or
restate the goals at the beginning of
the meeting but rather than thinking
about these
as simply tasks to be achieved it
actually helps us
realign at an increasingly fine scale
with the work that needs to be done and
that is a full stack
alignment from the syntactic level the
details and the nitty gritty of the work
that does need to get done
up through the narrative where are we at
this moment and what alphas does this
project relate to that's from the
systems engineering
framework all the way up to the highest
level which is

what is our group doing what is our
company doing what problems are we
trying to solve
and these ground floor or pet house
level questions depending on how you
think about it
are foundational and we return again and
again to these questions about what our
purpose is
and that's man's search for meaning
that's the whole task
and that doesn't really end so by
framing the problem not just as a set of
uh popping in tasks to the stack and
then finishing them
we can think about this as this ongoing
process of realignment
of the details of our sensory states to
our desired
regularities that we want to see in the
world
any other thoughts on this slide
cool all right so we are gonna have a
few
embodied affordance questions so here's
a quote from the paper
sports science provides an illustration
of this tendency towards optimal grip
studies of the dynamic interplay between
a boxer's stance and position
and the action possibilities available
to them as a function of stance and
position
have shown that punching bags afford
different kinds of strikes to boxers as
a function of the distance between
boxer and punching bag boxers tend to
move their bodies to an

optimal distance from the punching bag
specifically one that affords the
greatest variety
of strikes so not the most powerful not
the most rewarding
not the most um unexpected to the other
person in a theory of mind but actually
the greatest variety from an embodied
perspective of the actor
this is a case of moving towards optimal
grip so
it's beyond the grasping now we're
talking about stance
when observing a painting we also move
our bodies and our gazes in a way that
maximizes our grip on the scene or
details observed
we might call such dynamic adaptive
engagement with a field of affordances
in rolling cycles of action perception
skilled intentionality so the question
here is
what is being optimized by the body what
do
bodies do
it's a good question there's probably a
couple of answers
and we can maybe go reveal half of the
hint which is
how do predictions play into this
optimization
so what is the body optimizing um
doesn't need to be the end-all be-all
answer but what is something the body
optimizes and then
how do predictions play into this
optimization
well i mean a lot of my own work has

been about
since writing this paper has been about
figuring out exactly what this means
uh but you know if the free energy
principle
holds then it's not just brains that are
doing this free energy minimization
business
it's really every level of biological
self-organization
and there it really just means the same
thing across all levels which means that
you know
you have a desired distribution over
your data
and you know actions
acting on the environment is generated
such that
this preferred data distribution is
brought about what changes
at each scale is what the relevant data
is what the relevant set points are and
what the mechanisms are that bring about
the preferred data distribution right so
uh you know from a cellular point of
view
uh you know the markov blanket of a cell
is effectively
its membrane uh where the sensory states
are
all the receptor surfaces of the cell
and the active
states are like the cytoskeleton and
other
mechanisms that allow the cell to move
about and to
effectively generate the patterns of
data that it expects to encounter

but this very same story can be told at basically every level of like i was saying a biological self-organization where like from a from an interoceptive point of view uh what the body is doing is trying to keep some of its core homeostatic variables within uh viable ranges right so uh you know core body temperature might be maintained using such a mechanism where when you register a deviation from 36 and a half degrees you know you might have like a kind of low level uh autonomic kind of reaction you know you'll start shivering and you'll get goosebumps and so on uh but you know for uh animals like ourselves that have repertoires of cultural behavior well you might put on a parka you might turn your air conditioning on there are a bunch of other solutions to the same issue and i mean i think this is really the strength of the free energy principle i alluded to this last week uh i i at the time of writing this i sort of had an idea how this would work but it took the following four years of my life to really figure out what what that entailed you know along with our our collaborators uh carl fristen and everyone else if i can add one point on this into how the predictions play into the optimization so there's the classical control theory

thermostat

temperature drops and then the body can

engage in the

interoceptive affordances like

uncoupling in the mitochondria

or shivering or the behavioral and the

uh cultural affordances like wearing a

jacket

but how about if you were planning a

trip to somewhere that was cold

so there's nothing in your local

environment that's telling you that your

temperature is dropping in fact you

could be in

arizona sweating but you're packing for

somewhere that's cold

and so traditional theories of semiotics

might be challenged by that they could

say well you saw um

the cold city and you looked at the

website and so you made some sort of

connection

and you keep on doing this back

propagation of symbols which

is a valid account but also we can think

about what the body's doing

is optimizing its thermal range it's

minimizing surprise about excursions

from its

thermal range that's tolerable and then

it's

acting based upon its prediction and so

similarly

there's substances or states where your

body's prediction

it's now cast prediction or its future

prediction can be wrong

and so you can have people who are

shivering even though their body
temperature is high in the case of the
flu
or something like that so it's really
the interaction of the
prediction and the affordances not
necessarily the instantaneous however
the instantaneous
is often the most at hand um sasha
um this is quite funny but this metaphor
reminded me of um
just kind of the stereotypical like when
you go out your mother tells you to
bring a jacket
and that's because like the older
caretaker role thinks farther ahead
into future states and the
younger child role
um they're just thinking about right now
or where they're going
and they're not thinking to this
possible future state
they're not planning for that um
possibility so i just thought that was a
a nice
parallel to this whole uh jacket
thermoregulation um
how far ahead we're willing to plan to
reduce
um variability or uncertainty in our
environment
and differences potentially in risk
tolerance too
even say i i know it's going to be cold
mom i just don't want to risk losing my
jacket or i
want to be cold or something that or i
don't want to listen to you because

whatever oh sasha i think you're putting
your finger on
precisely like the core kind of
political economic issue that we're
dealing with
as a species right now which is that the
problems that we're dealing with
are occurring at a scale that
that is more or less invisible uh
to the generative models that we've
evolved
uh in our heads effectively right so uh
you know climate uh climate change
global warming
uh these are processes that unfold over
decades
and uh that require large scale
responses uh you know probably
to address this you know in in any way
decently like a kind of multinational
coordinated response
uh but our brains are meant to deal with
phenomena spanning from a few seconds
to you know maybe a year
in in more traditional societies when
you're like dealing with the season
cycle
and everything uh and so there's a huge
gap there
and you know i think the the disconnect
between the kind of scientific
consensus and uh you know at least in
the us the kind of lay consensus
on say climate change has to do with the
difference in temporal depth
of the models that we're looking at so
as scientists we we deal with these like
you know

models that in in the case of like
astrophysics allows us to
increase by like you know orders and
orders of magnitude for
like the the scope of our predictions
and uh
and so on but then there's this gap
you know uh between our our intuitive
understanding of the world and physics
and this also connects back to the world
that we live in
you know uh what kind of world we share
ultimately
um yeah so all of that stuff
yeah yes nice um stephen
yeah i think i agree with what's being
said and we've got into this
world at the moment where we we're very
focused on what we want
and how we can achieve that but
in our bodies might be when we're
putting on the coat or something we're
meeting our needs
as an organism so that the heat
and in complex adaptive systems for
development they're looking at like
need want and love to have happen
and like the want is what we might you
know and
we tend to focus on this instrumental
want but that can change
within like this construct but
our bodies we tune into this kind of
deeper feeling
of what we need like we can infer that
and connect to that
and i think that is um
that's partly what active inference sort

of helps us go back to
okay when we're starting to sit into
that because indigenous cultures look at
that thing of seven generations
and it's not like they plan for seven
generations but they
sense what's needed so they're kind of
working at that kind of you know organic
level
one one way also that the temporality
becomes embodied in our language in
active through language is
which words don't have plurals in
english now this isn't like a hard and
fast rule but oil
fish sheep there's a lot of words where
it's just like
it's air it's just there's no airs how
many errors are here it's just air is
around us
so implicitly one shouldn't have to
worry about depleting it right
well wrong because of course you can
overfish and so
it really bridges us to asking about
how could we construct our actions today
so that it influences the way that we
want the future to be
just to complete this slide the last
question is beyond physical grip
where does this idea of optimal grip
apply
and a really nice visualization was from
yelle broomberg's and eric wrightville's
paper on self-organization free energy
minimization
and optimal grip on a field of
affordances which was cited

in ramstead's paper and so here we see three representations of fields of affordances so these are like blocks that you could pick up they're blocks that you could sit on maybe blocks that you could carve a sculpture out of there are things that you can do stuff with and depending on the affordance it's gonna be a different one and the normal on the top in a we see that there's multiple different kinds of affordances of different types there's a bunch of yellow ones a bunch of red ones and a bunch of blue ones and maybe they have differential salience or they're adequate for different uses like a taller or a shorter person then in b and c we see two sort of hyperbolic and simplified but still didactic representations of alternate states so in b which is depression all of the field of affordances look less interesting they all look relatively drab and uh sort of not uh fun and in ocd we see that there's a massive highlighting of one of these affordances at almost the expense of completely blocking the perception of all of these other affordances and so i find this to be a really helpful framework because it helps highlight the

commonalities between
for example different people who are in
a depressed state or different
representations of
obsession on different topics it could
be rumination
in thought it could be a specific motor
behavior
but when we think about it like this we
can actually
go from this representation backwards
and say okay well
could we change our perspective so that
yes the red square is still really big
but also we can see at least just for
now just
see these other affordances or could we
make one of those gray boxes a little
bit more exciting
so i i just like this as a
representation and
it shows that this grip concept and the
affordance idea
is really transferable to different
areas springberg
eric grievell julian kaverstein don't
necessarily get enough credit for this
whole skilled intentionality framework
that they've developed
uh you know i i think it's it's some of
the connective tissue that really
allowed us to
build the cultural affordances framework
i really really like the way that they
connect
the kind of inactive emphasis on you
know action orientation
and embodied kind of coping with an

environment
with the ecological psychology
literature
on affordances and so on i mean uh we
were discussing this last week
their use of affordance is also not i
think
consonant with the kind of classical
gibsonian sense of affordance
uh where you know and i i mean the
common sense and
use of affordance i think in in academia
at least
also doesn't really resonate with what
gibson meant
recall that what gibson meant really was
that uh all the
information that an organism needs to
interact with its environment
is already available in its sensory
array
as it moves through its environment such
that
you know an internal reconstruction of
the environment
wasn't all that necessary because you
could just directly
interact with the environment so a lot
of these commitments are kind of falling
out
especially like in our use of the
of the construct in the free energy
principle framework like
we're explicitly reintroducing notions
like affordance
sorry like uh inference uh you know like
reconstructing the set of causes that
led to

the the sensory stimulation uh
but i think uh you know some of the
people who make
uh the the concept palatable and usable
in things
like uh sports sciences and so on are
these people pushing the skill
intentionality
framework like it's a it's a very useful
framework
especially i don't know like uh stephen
i'd be curious to hear what you think
about this
as someone who's coming more at this
from the practice and
i think the concepts there are more
readily
usable than some of the more like formal
stuff coming from
active inference and so on and yet it's
all kind of part of the same
conceptual framework
yeah i i agree i mean this is where it
gets complex because you've got
quite a radical form of affordances
being thought of and
intentionality and you've got obviously
the active inference
complexity but they're like they talk
about social
practice theory in um as as quite a
in environmental science work now a lot
so they're trying to look at things
through
what's the social practices and the
social norms that
kind of align people i think it's the
same

kind of thing that you're talking about
here it gives
it gives an actual plausible mechanism
so it doesn't jump from like this is
what's happening in my body
and then suddenly there's some
theoretical framework which is
somehow completely separate it's like
okay it makes sense that i'm
i'm enacting putting my coat on i've
been acting looking at this
painting and trying to understand it i'm
now enacting looking at this painting
like i used to work in them in graphic
reproduction so when you look
you know you're not looking at the
painting anymore you're looking for the
colors in it or is it going to reproduce
or i'm going to zoom in on it with
photoshop so it's like
the practice of engaging with the
environment and then like what's that
fitted into if it's going to go into a
publication so what's the intentionality
what's the kind of scales
of practices um
and i think that i think what you're
saying it makes sense i don't know if
the two
have quite managed to join up yet to
make it fully practice
ready because to try and explain it but
i think what maybe that's where it just
has to be used
in practical applications in like an
activity
and maybe the people doing the activity
don't need to know all the stuff that's

going on behind the
under the hood if that makes sense um so
yep nice well i want to come to some of
the other
ideas in our other sections so this is
the
section on ontologies and culture so
here are two
nice representations one of people
working together all around the world
and the other one
of the skeleton that we all know and
love of active inference
the question is how do ontologies relate
to culture
and then if a person wanted to address
that question with a raised hand or the
second question
is what kinds of tools are needed to
work
with ontologies and narratives
so i'll start with one of language as an
ontology scheme for the regularities and
the objects that are around us
so it's one of the cliches that x
culture has
y words for z concept you know oh this
culture has 22 ways to describe this
and of course you can always add
adjectives to describe a variety of
different kinds of
snow or sand or whatever it is but it
becomes
deeper through time and there becomes
specialized words that really capture
and can finesse as an as a handleable as
a grippable concept different types of
sand snow whatever it happens to be

trees and then

can then be built into larger systems or sentences

maxwell um

yeah so i mean i i think what you just said was very interesting

um i just want to clarify a bit what we meant by

uh ontologies in this context um so

i mean we're drawing on uh the ontological turn material uh that's coming out of

anthropology in large part uh apologies to steven

uh some of this material is delusion uh

i mean you know i i also tend to share your your reservations about

uh some of the uh stuff coming out of delos and stuff but

uh so the ontological turn fundamentally is saying something like well

what we should be doing is talking about the worlds in which people inhabit

in the way that they inhabit it right

rather than just saying what we have

as scientists as knowledge and what other people have is mere belief

uh what we should be doing is talking about like the really the worlds in

which people live

and and the ontologies or like the categories of

types of being that people are kind of bringing on board

um so you know we we don't

i think we're at odds with some of the people in the um

the ontological term because we take a

meta-bayesian approach to this which is
to say well we all
operate with beliefs the scientific
worldview is also a set of beliefs that
comes with its own ontology and whatever
so as long as it's an even playing field
and there's no like kind of
disparity uh between you know the the
respect with
to that we're according to different
kinds of knowledges
then you know we're sort of all
operating with our ontologies
and in the context here in this paper
what we mean more specifically
is things like uh like ontologies
of persons right like so uh
for those of us in the american and
particularly north american context you
know racism and colorism our major
issues
and you know it just is the case that
young children even even children
that hail from minority groups
internalize
the categories of personhood uh of the
dominant
uh political group right so you know the
the well-known
you know uh experiments on children
suggesting that
you know children from racialized
communities tend to prefer uh
dolls for example that look like white
people or whatever or dolls that
uh are consonant with their own kind of
ethnic
uh you know outward appearance

uh and so the yeah what what we're
talking about then is like
the categories of p you know uh
different kinds of people uh different
kinds of places
uh all of these kind of concepts and
norms and frameworks that
so all this is an ontology it's it's a
way of cutting up your sensory stream
effectively of parsing your
phenomenology
and what we're suggesting is that this
is acquired mainly through immersive
participation in practices
so you don't have to be explicitly told
for example
the first time you go to a fancy
restaurant not to jump up and down
on the the tables
humans are exquisitely sensitive to
context
and we're able to just dynamically
extract
the norms of the situation well at least
the
most neurotypical humans are able to do
this on the fly
without instruction and so this is where
the ontology stuff at least comes in in
this paper
yeah alejandra
yeah um well actually i was
thinking about this um idea about the
landscape affordances
and the shared attention
and how this shared attention is
determining which expectations
will be encoded in in this hierarchical

predictive uh
processing so i was wondering
and also in terms of these scientologies
related to
culture um about social networks
um it's kind of weird but yeah this is
idiot it's very um
like um importantly in my mind how these
um new ways of interacting
um are changing it's
it's a kind of a massive joint attention
uh i i don't know how to to to say it
but yeah
yeah like a massive joint attention a
kind of
extended culture that is manipulating us
um to what to attend so
um in in in some ways is changing um
what we believe that is relevant
and um manipulating us um
our attention and what to expect but in
a massive
way um and i think but we are not
conscious about this um um the
the effect that is having on
on our um beliefs and
as as you were saying maxwell uh this
um racism and social
preferences i think that um
i don't i don't know if i'm explaining
myself but this this
massive joint attention um
is is kind of affecting our our grip
the way we grip with society
in um unexpected ways i don't know yeah
that makes total sense to me
uh i really think that this is like a
really key issue

uh i mean you know the it's unfortunate
but

uh you know prejudicial behavior and
like hate crimes have been on the rise
steadily since

you know the trump administration took
power

and uh i mean i think it speaks to the
the power of uh you know

the power of the of the media
effectively and of shared images and
shared narratives

you know um i mean personally i think a
lot of the issue comes from the fact
that

uh you know older generations uh
weren't taught to be information skeptic
right

uh so i mean you know uh growing up now
as a 30-something when i was a kid and
internet was like starting out

you know what you learned was don't
trust everything you read on the
internet

so there's the i think one of the the
issues

is that uh

communities form around different types
of epistemic

standards effectively um

and i mean if you sort of look at the
split

the cultural split the the increasingly
deepening

kind of cultural split uh between you

know more conservative and more

uh liberal uh populations in the u.s

for example uh i think what you see is

really a split in
modes of adequate justification
um you know so it's it's stunning to me
how the same kind of fact
base can lead to completely opposite
interpretations uh in this kind of
context
but unfortunately you know the uh
the fact that you know uh you have like
these
very hateful groups that now spread uh
messages uh you know on a medium
that then is shared i mean it really
takes the
it takes the uh joseph henrik uh puts
this well uh in in his great book the
the secret of our success he says uh
essentially the same things that make us
capable of incredible feats of
coordination
you know and that have allowed us to you
know like build large-scale cities and
develop modern medicine
uh the same capacity for intense
cooperation
is also what makes us docile and what
makes us crave totalitarian rulers
uh we we just like to be very nice and
agreeable
you know and and if if everyone kind of
decides that some group is going to be
the scapegoat
well you know there's a tendency to go
along with that
and you know if we're if our hypothesis
is correct and
there's also a tendency to just acquire
unthinkingly

download the kind of information that's presented to us in immersive contexts then it's not surprising that you know if these messages are spreading online you know in very highly immersive kind of tunnel vision things like youtube channels that's just like re-link similar content i mean like yeah i don't know if this is an entirely coherent message but i do deeply resonate with what you were saying alejandra i think this is uh this is a huge set of issues that we need to address and it a lot of my own research for the next several years is going to be directed in precisely this direction uh like countering understanding and countering the spread of these narratives and like just modeling social dynamics using these tools more broadly yeah exactly as well sorry then just to add something um in terms of like your personal narrative your own story um how how is affected um i i can see it in in adolescence or in child's they are just expecting alike and that's meaning so meaningful to to um minimizing prediction error about yourself so uh yeah it's like you have like thousands and or hundreds of people just paying attention about your picture and and if you

receive 10 likes or 100 likes
um what this like
um the way that the self
is gripping to to this um
that's an interesting point there's a
weird kind of self-evidencing i guess
yeah it's very sad but i guess that it's
really tragic when you think about it
right we off sourced we off sourced our
sovereignty of attention to algorithms
and those are driven by exploit and in
our sense of self-worth
right so from an active inference point
of view
likes and retweets and shares and
whatever
are evidence for your existence right
for for a model of you
having worth in a social environment
uh that's actually i think worth at
least the paper if you want to write
that up alejandra
like send me a dm and we'll write that
up as a paper
it sounds it's a great idea um sasha
uh yeah i just wanted to add on to that
um
a comment about uh don't trust the
internet
um and i i really like the phrase that
you used
alejandra uh the massive joint attention
um and then it seems like social media
um is kind of able to bypass that filter
of
don't trust the internet but these are
my friends
this isn't the internet these um the

social network somehow
is able to pass through that filter that
this is a group of people that we trust
that we feel that we trust
more than the internet as this um
greater entity
well they don't call it anti-social
media they don't call it physical
distancing they say social distance
these words really matter as far as
slipping in the back door
semantically with what um is
uh being communicated in the end
so yes um to pick up on what sasha was
saying i think uh
the internet's epistemic status is very
much up in the air like you know i
i think i'm i'm always a bit naively
optimistic
uh and so i like to think that people
are critical about what they read on the
internet but
i i think it was a joke somewhere i
forget
who said it but it's something like you
know when i was growing up my mother
told me don't get into a car with
strangers
and like now i literally summon
strangers
from my internet having mobile device
to pick me up and to bring new places
right
uh like the uh i
i i mean as some of you are pointing out
it's it's sort of the connective tissue
of a lot of relations especially post
coval a lot of my friendships for

example have been pursued over the internet

uh so that's actually a very interesting point maybe the fact that this information is being presented through our friends that are sharing this

uh information does tend to increase its precision for us

and make it more salient 100 it's why the word of mouth recommendation of a product

is the highest form it's why influencers try to be personable

there's so much to that angle and using the pre-existing and the possible social networks

to leverage uh the spreading of memes that might be just

purely corrosive not even for or against one side just pure disinformation or pure things that uh are divisive let me just

run through a few of our latter questions so we can

get some more neurons firing connect some more ideas and stabilize these concepts

so in this quote which um

i will not read it's just about how cultural affordances are acquired through cultural practices

and i found a really nice image that i thought tied together

ontologies with culture so this is an image from the ontotext.com on the bottom there

and it writes ontologies do not only

introduce a shareable and reusable knowledge representation but can also add new knowledge about the domain so i had a few questions here about ontologies and that could be in that anthropological term the the anthropology or the ontological turn as maxwell was describing it or in the more computational sense like gene ontologies the structure of words that we use to describe genes so ontology in that sense means like a formal knowledge structure and about the attention and relationship between attention and ontology between ontologies and development and then especially for this group how do ontologies and active inference interact and how does that actually add knowledge to a situation rather than being a pure descriptor well i mean in part you don't grasp the world in a naked way right like we grasp it through our conceptual categories uh through the salience structures that are encoded in our conceptual categories in large part and uh yeah i mean the uh the point of having an ontology is to kind of you know fix the frame problem in a sense to kind of say okay what out of all these elements that could be monopolizing my attention at a given moment

you know who are the relevant
interlocutors that are deserving of my
attention
what are the salient aspects of the
scene that require
my processing yeah and in general
what you know just in in the broad sense
right
what uh is kind of the
otherwise you're just you know dealing
with a sensory stream
which is all fun and good but like so
the ontology is what you bring to bear
on the sensory stream to kind of parse
things out
from it yep and i i
like where you took that it was exactly
what i was thinking about like
how many tennis balls can you fit in
your hand maybe three maybe you have a
large hand four tennis balls or
something but
you have a bag and you can cluster the
tennis balls and you could pick up a bag
with 100 tennis balls
so that's the grip physical metaphor is
the tools the affordances that cluster
allow you to grip more and so similarly
we can use
ontologies of ideas and use that
to get a grip on really high level
concepts
and then we'll pick up the double edged
sword though right because
uh i mean you know this is the problem
of using priors in general is that you
know you're starting from a point that
might not be the right one

and that might actually prevent you from
uh you know so
ontologies are the reason why we have
you know the capacity to interact with
each other for example as peers
you know because we have these
categories uh you are my peers and my
friends and we can interact
but you know ontologies are also
responsible for racist behavior
uh for sexism uh you know if i if my
ontology says women are inferior uh
to men or the people with dark skin but
i don't
believe any of that just to be
clear but like if that is my
ontology then i'm bound to treat
uh others in a way that reflects those
beliefs uh
so i mean this is why they're so
important to culture i think
because they kind of it's it's not
unrelated to this formal definition that
you were discussing dan like
it gives you the kind of general tracts
that
that you'll follow in your treatment of
others uh so then it becomes
interesting to you know use
anthropological methods for example
to see what are the ontologies that
people are actually bringing to bear
uh you know and to use that physical
carrying metaphor so you go okay well i
can hold three eggs in my hand
i should have a bag with 500 eggs
because i can pick it up well now you
have none or you have a mess

and so over simplification through any
number of mechanisms
within a value structure simply leads to
false positives and false negatives
and another way that we can bring in the
free energy principle
is the phrasing of free energy as model
accuracy
penalized by complexity and in some ways
the extreme
uh simple model let's say a one category
model like all people are people
or two or three category model and so
do the models get more accurate in terms
of reducing surprise as we include
more and more parameters and then in the
extreme case
we have everyone as unique which is an
interesting
bookshelf you know end piece to contrast
with this everybody's the same
and so somewhere in between is where
we're actually going to end up
and so how do we help bring these
implicit
generative models of the world and our
priors in into
being shaped and designed and uh
personally i think we never escape
the value question and it's one thing
to come together with somebody who has a
different perspective and it's why it's
so important to highlight coming
together across differences
because there's just no amount of
cultural or genetic or whatever
homogeneity
that is going to prevent conflict and so

instead we can have a coming together
story about working together across
differences
rather than wedging tiny differences and
that story itself is in a trade-off with
essentially
gaslighting and telling people that
suffering is just part of how things are
so there's so many trade-offs that exist
um especially when we're talking about
issues that matter most i mean we're not
talking about species definitions in
foxes we're talking about
our own lives and our world in our
ecosystem and so
yes they're touchy topics and it's so
important to be
clear about what is and also what we
want to see
and so i i think there's so much that
can be explored
and expanded on here i i like that you
brought in complexity minus accuracy
uh because in a sense the you know
this is this is a little bit a slightly
looser way of thinking about it but
in general you want as little boxes as
possible right
uh boxes i mean if you listen to
buddhist wisdom for example boxes are
the source of suffering right we
we cling to our boxes we put things in
boxes and
and it makes it it makes us feel
comfortable uh
but really this is the source of
suffering uh you know
the this tendency to only be comfortable

with things when they're
nicely packaged in these kind of nice
binary categories
and this is why people uh get so like
uncomfortable uh when these binaries get
challenged
you know by like gender fluidity and
transness
and these kinds of things on the flip
side if we didn't have boxes at all we
would die
right uh you know you need to be able to
parse that sensation as hunger
uh in order to generate like you know
appropriate
survival generating uh behavior
so i think that's like the the the
fundamental paradox of human existence
is both that we need
uh boxes to survive but that they also
are
the source of much you know just
suffering
uh both that were inflicting to
ourselves by
forcing these kind of conceptual
straitjackets onto ourselves
but also because of the social wrongs
that that can be uh
the result of you know using the wrong
ontology
yep and cognitive diversity means
differences in risk tolerance
it means of course different
perspectives unique perspectives
and also um maybe on that explore
exploit or on that model accuracy versus
complexity

some people will accept a more accurate
model from their perspective
that is also a lot more complex whereas
somebody else is
the coefficient of how much they're
penalizing a complex model
is different so they're going to come to
a different conclusion and it's not
as simple as right versus wrong in that
situation it's
model preference and could we take a
step back could we reframe
and understand how we're all part of the
ensemble that's learning
it would be nice let's go through a few
of these
last topics on communication and
semiosis or meaning generation
so here is from figure one of the paper
which we discussed last time as far as
what it was describing but here i want
to connect it to one of the quotes
they write conventions need not be
explicitly
formulated as rules and may instead
originate in the actor's engagement with
local backgrounds over time
that is from non-contentful
developmental experiences
learning or participation in social and
cultural practices
to operate with conventional affordances
agents must have shared sets of
expectations
we must know what others expect us to
expect so
yes we're hitting the same idea from
figure one but

there's a lot we're kind of spiraling
another layer and coming back to this
idea where
we're in a remote group right now and so
we are getting our expectations about
what to expect and all these things
not only are they going to be very
different person to person because our
local worlds are
quite different but there's this
algorithmic level
that influences what we expect and it
changes
on the inputs that we receive and
reducing our surprise about the sensory
inputs
is one of the drivers of behavior and so
how does
what we see in the news feed end up
recapitulating the things that we
will draw our attention to and that's
sort of this self-radicalization
process that can happen is attention is
brought to something
and then it continues to basically
percolate upwards
as people pay more and more attention
and are more and more cautious or more
and more aware
of something that it's hard to take a
step back from any scientific worldview
and say that one is better or the other
because although i'd love to hear a
workaround
it's hard to get a purely secular
perspective because someone could say
i'm just pursuing my accuracy by doing
something that's aberrant to you

and it's difficult to see
purely within the framework nor does
there have to be
i think what free energy and active
inference and focusing on the
communication dynamics allows us to do
is let our values be values let our
communication be communication let our
behavior
be a behavior and uh talk about what we
expect
using ways that we want to communicate
and if people don't want to use
non-violent communication
other people may or may not choose to
communicate with them but it's um
it change how we communicate what we
expect when
there's not only that computational
intervention but also
a lot of um you know coming together of
different perspectives it's
one of the challenges of our day and
then also just
this cultural iceberg was a nice image
some of the
the words are small in there but the
quote on here is
joint attention is usually understood as
occurring in a diet of two people or
between
agents in direct interactional spheres
of communication gaze following
finger-pointing or other verbal or
non-verbal cues
to address more complex social
situations it is useful to revise
current sociocognitive models of joint

attention to encompass
fundamentally triadic situations in
which the third
is the socially constituted niche of
affordances supported by local
ontologies and abilities
so before you have the questions just
some of these in on the iceberg above
the iceberg we have dress
folklore language fine arts holidays
food
and below we have family values we have
gender roles
um pride concept of justice so this
and there's different versions of the
iceberg i just thought this one had a
lot of nice ideas
my questions here were what is the third
member
of a social interaction and how can
active inference help us design culture
if i can jump in quickly of it's sort of
that
so a human once in culture it is never
alone
in a sense uh we're always constantly
seeing things through the eyes of others
it starts off as what would mommy want
me to do in this situation and it it
just gen i mean
it's it's remarkable canadians are
really good at this uh we
uh there were i i'd have to pull up the
studies but there are
there are studies that show that
canadians even at night
you know when there are no policemen
will stop at a red light and just wait

it out

you know and and go even if there are no

agents that could penalize them

uh i mean it's really remarkable the

extent to which we we internalize

uh the these kinds of norms and they end

up then directing

uh the way that we interact with uh with

others

yeah any other thoughts on this or we

can just it's it's something for

pondering

because we have the explicative portion

of our experience which is

that which comes to our conscious

awareness that which is stated verbally

but on the input side subconscious input

and on the generative models that can't

be expressed in words

sometimes they can be expressed through

art or through action

but the whole iceberg is there and so

it's just i really like that once you're

in cultured you're never alone and

indeed we are never alone because to be

aware

is to be communicating it's almost like

the uh cog ergo sum

it's like i communicate therefore we

exist

something like that um in this last

page on um semiosis which i think is an

interesting

topic the authors wrote however

cultural symbols and signs are usually

poly semis that means having many or

multiple meanings

and their interpretation depends on

context it's contextual

our hypothesis is that feedback loops
mediating shared attention and shared
intentionality are the principal
mechanism

whereby cultural especially conventional
affordances are acquired

so conventional is fun i think in this
setting because it alludes to
conventional like normal

like it's just the conventional approach

it's the normal approach but also

based upon convention and convention in

that sense means both the coming

together like it's a convention of the

knife sharpeners or something

but also it's a joint it's agreed

upon so it has many senses

and it shows how normal and agreed upon
often

are related that's one symbol and so

here was a

few graphics that i found on a nice the
dsproject.com

so on the left is what is known as the
piercing

triad by c.s pierce and that's the sign

the interpretant and the object and on

the right is

um sarsour who is another semitician

though may have used different words and

here we see that there's the signifier

and the signified and then the referent

and so i like this because it shows that

there's multiple

ontologies even when we're talking about

sense making about meaning generation

there's multiple ways we can parse that

and do we choose to see it as a triad
or do we choose to see it as a binary
with another binary like the sr and the
sd
are one whole and then the frf is
another
you know is it a pair of binaries or is
it just an integral
triad so that's one way where there's
multiple frameworks we can use just to
talk about meaning generation
and then this image is a representation
by those authors about
unlimited semiosis and this is sort of
looking
through time how that triad evolves and
so i just thought
how does this relate to active inference
to that unfolding
idea to this notion that our meaning
generation
our sense-making process is ongoing with
respect to the symbols
but we can modify those symbols as well
so
how can we maybe use active inference at
least to grasp what are the outlines of
this problem
some of it i think was addressed in uh
later
papers for example uh i forget which
active stream but when we talked about
is the free energy principle a formal
theory of semantics
and it relates to that um
i think it would probably be cool at
almost the end of our time
just to switch into

panel mode and just see if anyone had
any last thoughts that they wanted to
talk about
any last questions or where
are you going with what we've discussed
today what do you think would be a
good question to leave people on or a
good next step for us to take
i thought this was a really interesting
and useful
exercise i'm super grateful
to all of you for this continued uh this
continued attention
uh to the work that we've done uh yeah
i i told the i told the team earlier but
i've
i've blocked off these uh mornings to
participate regularly
um so i mean i i i
i'm very satisfied with this i think
it's very cool
and uh i think we've raised a lot of
issues that could be pursued as papers
uh yeah particularly this uh
issue that you were raising alejandra
about uh
social networks and the way that they
tie into
uh attentional dynamics i think
uh should be the target of much more
uh discussion uh
yeah i mean i i thought everyone had
very very interesting
uh things to say so thank you here's one
angle on that
is an evolutionary prior is that your
social networks
are constituted of people who are not

adversarial with you
most of the people who you're engaged
with whether or not they believe the
exact same thing
you have to be relaxed enough around
them to sleep
for example and so the the deepest prior
is
if we're relating then we're going to be
to some broad extent in agreement okay
now we know from
active inference and niche construction
that the two ways that
you can um deal with uncertainty or with
unexpected sensory inputs
is by changing the environment that's
niche construction or
by changing your internal model that's
inference so what happens
when you're in a social contrived
situation it says oh but your friend
posted this
okay so it's coming in through the
friend door and it's uh
something that for my current value
system at time t
i think of is not chill there's all
kinds of
things not being chill but it just isn't
and the two ways that i can
deal with that are change my model to be
more like that
so i make that less surprising some type
of hateful speech
i get numb to it or i come to embody it
actually to reduce my surprise on the
inference
side or i can get doubled down

in my beliefs and try to reduce my
uncertainty or the surprise from those
sensory cues
by getting into a debate i mean think
about how much mental energy
gets burnt on these sort of
debates and people trying to change each
other's mind online
and there's something beautiful about
people sharing their differences in idea
but when that whole discussion is being
basically
overviewed by an algorithm that doesn't
exactly have those people's mental
health or the
alignment of different cultures in mind
not exactly
yeah not not exactly you know it's
correlated no
it it can get off the rails and so that
deep evolutionary prior isn't something
that can just be unlearned like you were
speaking about earlier maxwell
and so when our deepest prior and our
intergenerational ones
get taken advantage of it's like these
slow waves
and they can't quite cope with these
very fast frequencies and so the options
are to slow it down that's like the rko
futurist you know oh well in the future
we'll return to
a more heterogeneous uh
organization of people so slow down
these high frequency waves
or come to a better way that we can
predict and communicate about them
i mean there's a lot more that we could

think about there but it just
it's uh yeah there's yeah
uh steven and then alejandra well i
think
like you were saying these ways that we
think about what is happening
um is really important in terms of how
we construct our
conversations but i think there's also
the where and
the where we've taken our attention
to be able to have our conversations so
when you're
locked in they call it context collapse
when you're on like facebook because you
have no context you have
no story you have no so one of the
things with complex adaptive systems the
workers cynthia kurtz and david snowden
they work with
um stories for small micro stories so
you fight
a hundred a thousand and start to
make sense of them like attribute them
with
um titles and metadata and actually
work with story because when you're
working with story
each story if it's kept raw so it's not
an opinion
they call it it's like it's kind of
maintains the internal complexity of it
so you can still reinterpret it but once
you've got an opinion
an opinion is collapsed down you can't
really reinterpret an opinion
so we're kind of trapped in these kind
of interpretive kind of

inductive deductive spaces which is
great if it's a case of like i have the
right answer or you have the right
answer

so then we can like debate it but the
trouble is what if we don't know
and no one wants to admit they don't
know um

can we work with something more complex
and

maybe that's a rare question as a how
question not a what question

a knife alejandra

um yeah and

and i i totally agree and also taking
what uh

sasha said that the idea

that we are talking with our friends and
and

yeah that's like the

like our conscious um

internal belief they are our friends but
we

actually not have any idea

how these platforms work

um how these platforms are manipulating
our attention so we are communicating
with others

but we have something

between this communication that we don't
know actually how is working

and um i don't know i i was

thinking that um they are kind of
playing

in terms of active inference to

constantly increase

our prediction never um and you have to
engage with with this platform in order

to
reduce or minimize prediction error
in terms of emotions or um
the way i react to other um giving a
like
or um sharing a post so um
i don't know this idea that we are
interacting
with friends or we're um close
people but we don't know what's
happening in between
sasha um
yeah i think that that's a great point
um to touch on that
we don't really understand what's
happening
uh this hidden state between um
us and the person that we're trying to
communicate with and how that's being
manipulated
and while that's that's a totally valid
critique of social
media platforms um we could say the same
thing for
like telephone communication or even
telegram
um the people who were sending the
message shouldn't necessarily understand
what was happening um
in in that process and yet
uh they felt or they they would trust
that it was um the integrity of the
message was maintained
or that it wasn't uh nothing was lost in
that process
or manipulated but so it's been kind of
like a steady
a steady march of uh

technol technological advances that are
more and more removed
from what we think we're doing in this
process of communication
yep the neutral claim is that context
directs attention so if there's a bunch
of arrows pointing to something even
though it's not in the exact object it
will draw your attention to it or
something that's placed on top of a
pyramid is what your attention is
supposed to be drawn towards
well in the telegram case not the app
the
the morse code version the context is
that you trust that western union or
whomever
isn't going to be garbling your message
and the context is who it's from
and so you can actually unpack it as if
that person were just telling you
but when it's a stranger and it's like
engaging
the gas and the break on is this an
adversary or not
and the context is constrained so you
can only see
their name or a fake internet name and a
picture
and then is it a surprise that given
a picture and what somebody says
somebody's gonna unpack it
in potentially the least charitable way
based upon identity or
based upon a typo based upon some sort
whereas in person we just look at each
other
and we can have more of this human sense

we can have the context
and that helps us understand and then
beyond the context
cues when we can actually feel like
we're part of a shared society
then we want to help people who are in
need in our local worlds
and there's only one world so it is our
local world
spaceship earth so there's there's so
many angles and the question would just
be
what are those on ramps and then how is
it all
going to actually play out any
final thoughts though because this was
an awesome discussion great
follow-up and there's we could have
talked for
longer on just four point two three four
five but next week we'll move to five
and so if there's no further thoughts
i really appreciate everybody coming
together thanks for
the attention to those participants and
also to our listeners
we'll always be even weeks later
accepting
input on what papers what topics what
questions would be cool to discuss
and so yes thank you it's been a great
stream